

Role-Channel Anisotropy of DBP Surfaces

An exact local invariant for output-role curvature,
and its thermodynamic geometry

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Abstract

On a constraint surface $F(D, S, P) = 0$ carrying a directive–substrate–product (DBP) role assignment, the coupling curvature one measures depends on which variable is treated as the output. We make this dependence a single exact local invariant, the *coupling anisotropy* $\mathcal{A}_c$, equal to the squared norm of the projection of the three role-output coupling curvatures onto the standard 2-dimensional irrep of S_3 . We give three equivalent closed forms (statistical, representation theoretic, and an explicit jet form), prove the dimension law $\dim \mathcal{Z}_\Sigma = 2 - r$ for the role-isotropy locus $\mathcal{Z}_\Sigma = \{\mathcal{A}_c = 0\}$, and compute $\mathcal{A}_c$ on three reference surfaces (bilinear, ruled, generic). $\mathcal{A}_c$ has poles on the role-singular set; we resolve them with a projective normal on which S_3 acts by coordinate permutation and a homogeneous gap vector that is polynomial in the 2-jet, restoring global exact- $\mathbb{Q}$ evaluation. Finally we connect the construction to thermodynamic geometry: the role-output charts are the thermodynamic representations, the role transposition is the Legendre swap, and $\mathcal{A}_c$ is the norm of precisely the off-diagonal coupling that Legendre-invariant geometrothermodynamics removes. For the Kerr black hole, the Davies point of heat capacity sits exactly on the isotropy branch where the Weinhold and Ruppeiner coupling curvatures coincide, and the role-singular loci reproduce the physical boundaries (extremal and Schwarzschild) for free. The companion paper treats the global (branched-cover) side; the present paper is its local-geometry counterpart, sharing the standard S_3 irrep as the object both theories see.

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1 Introduction

A DBP surface is an algebraic relation $F(D, S, P) = 0$ with a role assignment: D directive, S substrate, P product. Solving for any one variable presents the surface as a graph over the other two, and the local second-order geometry of that graph—in particular the off-diagonal (coupling) curvature—generically differs from chart to chart. “Which variable is the output” is not supposed to matter for the surface, yet it changes the measured coupling. This paper makes that discrepancy a single invariant and studies its zero locus.

The same discrepancy is the long-standing difficulty of thermodynamic geometry: the Weinhold (energy-Hessian) and Ruppeiner (entropy-Hessian) metrics are conformally related by the inverse temperature but are not invariant under Legendre transformations, and the resulting representation-dependence yields contradictory black-hole results.[[literature](#)] Quevedo’s geometrothermodynamics responds by building Legendre-*invariant* metrics, which removes the off-diagonal coupling that varies. The DBP stance is the opposite: *measure* the variation. The invariant $\mathcal{A}_c$ below is the squared norm of exactly the part a Legendre-invariant construction discards.

Notation. We work in a graph chart $P = f(D, S)$ with the finite 2-jet

$$(a, b, A, B, C) = (f_D, f_S, f_{DD}, f_{DS}, f_{SS}), \quad q_0 = 1 + a^2 + b^2.$$

Here A, B, C are Hessian entries of the local graph; they are unrelated to the seam quantities of the same letters in the monodromy companion. The *regular locus* is $ab \neq 0$ (both first derivatives nonzero); the *role-singular set* is $\{a = 0\} \cup \{b = 0\}$. The active role transposition generating these output-role charts, the regular-locus conditions, and the orbit framework on which $\mathcal{A}_c$ is built are established in the foundational paper, W. Lloyd, *The Active Role-Jet Orbit Calculus for Constraint-Surface Roles*; the present paper develops the exact $\mathcal{A}_c$ structure and its thermodynamic-geometry reading.

Certainty tiers. [[proven](#)] (exact algebraic fact), [[symbolic](#)] (verified symbolically), [[computer-verified](#)] (verified by exact- $\mathbb{Q}$ computer algebra on a specific surface), [[conjecture](#)], [[interpretation](#)], and [[literature](#)] (established in the cited literature, not a contribution here).

2 Coupling channels and the anisotropy invariant

Definition 1 (Role-output coupling channels). For each output role the coupling channel is the mixed second derivative of that role’s graph, expressed in the original chart’s jet:

$$\Lambda_P = B, \quad \Lambda_D = \frac{Ab - aB}{a}, \quad \Lambda_S = \frac{Ca - bB}{b},$$

with coupling curvatures $\kappa_{c,i} = -\Lambda_i^2/q_0^2$ for $i \in \{P, D, S\}$.

$\Lambda_P = f_{DS}$ is the coupling of the P -graph directly; Λ_D and Λ_S are the corresponding mixed partials of the D - and S -graphs, obtained by inverting f in the relevant variable, hence the $1/a$, $1/b$ from the chart Jacobian.

Definition 2 (Coupling anisotropy).

$$\mathcal{A}_c = (\kappa_{c,P} - \kappa_{c,D})^2 + (\kappa_{c,D} - \kappa_{c,S})^2 + (\kappa_{c,S} - \kappa_{c,P})^2$$

the simplest non-constant S_3 -symmetric function of the three channel curvatures that detects collapse of the role orbit.

Computational Result 1 (Three equivalent forms). With $e_1 = \sum_i \kappa_{c,i}$, $e_2 = \sum_{i < j} \kappa_{c,i} \kappa_{c,j}$:

1. **Statistical:** $\mathcal{A}_c = 2(e_1^2 - 3e_2) = 9 \text{ Var}(\kappa_{c,P}, \kappa_{c,D}, \kappa_{c,S})$.
2. **Representation-theoretic:** under $\mathbb{R}^3 = \mathbf{1} \oplus \mathbf{2}$ of S_3 (trivial $\oplus$ standard),

$$\mathcal{A}_c = 3 \|P_2 \kappa_c\|^2,$$

the squared norm of the standard-irrep projection; the $\mathbf{1}$ part is the role-blind mean coupling e_1 .

3. **Explicit jet form:**

$$\mathcal{A}_c = \frac{D_{PD}^2 + D_{DS}^2 + D_{SP}^2}{q_0^4}, \quad \begin{aligned} D_{PD} &= \Lambda_P^2 - \Lambda_D^2 = \frac{Ab(2aB - Ab)}{a^2}, \\ D_{DS} &= \Lambda_D^2 - \Lambda_S^2 = \frac{(Ab^2 - Ca^2)(Ab^2 + Ca^2 - 2abB)}{a^2 b^2}, \\ D_{SP} &= \Lambda_S^2 - \Lambda_P^2 = \frac{Ca(Ca - 2bB)}{b^2}, \end{aligned}$$

with the telescoping identity $D_{PD} + D_{DS} + D_{SP} \equiv 0$.

All three are verified equal, symbolically and against the keystone value below. [symbolic]

The gap vector (D_{PD}, D_{DS}, D_{SP}) is just the cyclic sequence of *squared-channel differences* $\Lambda_i^2 - \Lambda_j^2$; its telescoping to zero is the statement that the gap vector has no $\mathbf{1}$ component, which is why the role distinction lives entirely in $\mathbf{2}$.

Computational Result 2 (Keystone value). For $F = D^2 + DS + P^2 - 3$ at $(1, 1, 1)$ the jet is $(a, b, A, B, C) = (-\frac{3}{2}, -\frac{1}{2}, -\frac{13}{4}, -\frac{5}{4}, -\frac{1}{4})$, giving $\kappa_{c,P} = -\frac{25}{196}$, $\kappa_{c,D} = -\frac{1}{441}$, $\kappa_{c,S} = -\frac{1}{49}$ and

$$\mathcal{A}_c = \frac{42793}{1555848} \neq 0,$$

reproduced by all three forms. [computer-verified]

3 The dimension law and the isotropy locus

Definition 3 (Role-isotropy locus). $\mathcal{Z}_\Sigma = \{p : \mathcal{A}_c(p) = 0\}$, the set where all three role frames agree on coupling curvature. Over $\mathbb{R}$, $\mathcal{A}_c = 0 \iff |\Lambda_P| = |\Lambda_D| = |\Lambda_S|$ (the q_0 factor never vanishes), so $\mathcal{Z}_\Sigma$ is cut out by the factored numerator branch by branch.

Proposition 1 (Linear channel map, $\det = -1$). *For fixed regular (a, b) , the map $(A, B, C) \mapsto (\Lambda_P, \Lambda_D, \Lambda_S)$ is linear with constant determinant:*

$$\frac{\partial(\Lambda_P, \Lambda_D, \Lambda_S)}{\partial(A, B, C)} = \begin{pmatrix} 0 & 1 & 0 \\ b/a & -1 & 0 \\ 0 & -1 & a/b \end{pmatrix}, \quad \det = -1.$$

[symbolic] Consequently κ_c is an invertible quadratic image of the second jet, and in second-jet space $\mathcal{Z}_\Sigma$ is a union of four lines through the origin (the sign branches $\Lambda_D = \pm \Lambda_P$, $\Lambda_S = \pm \Lambda_P$), of codimension 2.

Computational Result 3 (Sign branches).

$$\Lambda_P^2 = \Lambda_D^2 \iff A = 0 \text{ or } Ab = 2aB; \quad \Lambda_D^2 = \Lambda_S^2 \iff Ca^2 = Ab^2 \text{ or } Ab^2 + Ca^2 = 2abB.$$

[symbolic]

Computational Result 4 (Dimension law). The codimension of $\mathcal{Z}_\Sigma$ equals the rank of the standard-irrep content of the second fundamental form, $r = \text{rank } d(P_2\kappa_c) \in \{0, 1, 2\}$, and

$$\dim_p \mathcal{Z}_\Sigma = 2 - r$$

The bound $r \leq 2$ is sharp because $2 = \dim \mathbf{2}$: the telescoping reduces the three channels to two independent role-distinctions and *no further*. [symbolic]

The dimension law says the isotropy locus is large exactly when the surface is role-degenerate: $r = 0$ (flat coupling, all of Σ isotropic), $r = 1$ (one-parameter family of isotropy curves), $r = 2$ (isotropy on a codimension-2 set, generically isolated on a surface).

4 Reference surfaces

surface	type (r)	$\mathcal{A}_c$	$\mathcal{Z}_\Sigma$
$P = DS$	bilinear ($r=0$)	$\equiv 0$	all of Σ
$P = DS^2 + S$	ruled ($r=1$)	$\frac{32 D^2 S^6 (3DS + 2)^2}{(2DS + 1)^4 q_0^4}$	$\{D=0\} \cup \{DS = -\frac{2}{3}\}$
$D^2 + DS + P^2 - 3$	generic ($r=2$)	> 0 everywhere	$\emptyset$

All three rows are computer-verified. [computer-verified] Three points deserve comment.

Bilinear ($P = DS$). Here $A = C \equiv 0$, so $\Lambda_P = 1$, $\Lambda_D = \Lambda_S = -1$ identically and $\mathcal{A}_c \equiv 0$: multiplication is everywhere role-isotropic ($r = 0$). This is the local counterpart of the global fact that the bilinear product is the trivial DBP surface.

Ruled ($P = DS^2 + S$, **the directive enters linearly**, $f_{DD} \equiv 0$). The corrected expression carries a $(2DS+1)^4 = b^4$ in the denominator. The numerator zeros $\{D = 0\} \cup \{S = 0\} \cup \{DS = -\frac{2}{3}\}$ include $\{S = 0\}$, which coincides with the role-singular set $a = f_D = S^2 = 0$ (degenerate isotropy of a singular chart); the genuine isotropy with live gradient is the line $\{D = 0\}$ and the hyperbola $\{DS = -\frac{2}{3}\}$. Crucially, the b^4 denominator is a genuine *pole* of $\mathcal{A}_c$ on the role-singular hyperbola $b = 0$, i.e. $\{DS = -\frac{1}{2}\}$. The isotropy hyperbola $-\frac{2}{3}$ and the singular hyperbola $-\frac{1}{2}$ are the same shape with adjacent constants: $\mathcal{A}_c$ vanishes on one and diverges on the other, and they nest. “Isotropy tracks chart-singularity” is literal—one is a zero, its neighbour a pole. The pole is exactly why $\mathcal{A}_c$ in this form cannot be evaluated through the role-singular set, motivating §5.

Generic ($r = 2$). The keystone has $\mathcal{A}_c > 0$ everywhere in the physical chart; role-isotropy is empty. A generic DBP surface is nowhere role-isotropic.

5 Exact- $\mathbb{Q}$ evaluation through the role-singular set

The channel definitions and $\mathcal{A}_c$ carry a, b, q_0 in denominators; the first two vanish on the role-singular set and make $\mathcal{A}_c$ blow up there (§4). Both poles are chart artifacts, not geometry, and both are removable.

Proposition 2 (Projective normal; S_3 by coordinate permutation). *Embed the graph in $F = P - f(D, S) = 0$ and use the projective normal $g = [u : v : w] = [-a : -b : 1]$. The role transpositions act by pure coordinate permutation,*

$$s[u : v : w] = [v : u : w] \quad (D \leftrightarrow S), \quad t[u : v : w] = [w : v : u] \quad (D \leftrightarrow P),$$

globally linear and pole-free. Applying t to $[-a : -b : 1]$ gives $[1 : -b : -a]$, which recovers the affine transposition $(a, b) \mapsto (1/a, -b/a)$ on the regular locus and extends smoothly to $a = 0$ as the finite point $[1 : -b : 0]$. [symbolic]

Proposition 3 (Homogeneous gap vector). *Clearing the chart denominators by the weight a^2b^2 gives the polynomial gap vector*

$$\tilde{D}_{PD} = b^3A(2aB - Ab), \quad \tilde{D}_{DS} = (Ab^2 - Ca^2)(Ab^2 + Ca^2 - 2abB), \quad \tilde{D}_{SP} = a^3C(Ca - 2bB),$$

with $\tilde{D}_i = a^2b^2D_i$ and the exact telescoping $\tilde{D}_{PD} + \tilde{D}_{DS} + \tilde{D}_{SP} \equiv 0$. These are pristine polynomials in the 2-jet, so the isotropy ideal $I_{Z_\Sigma} = \langle \Lambda_P^2 - \Lambda_D^2, \Lambda_D^2 - \Lambda_S^2 \rangle$ is evaluated globally, including at a chart boundary, with no limits or exceptions. [symbolic]

Remark 1 (A coupling channel is not a bordered Hessian). It is tempting to replace Λ_k by the bordered-Hessian determinant $\det \begin{pmatrix} 0 & g_i & g_j \\ g_i & H_{ii} & H_{ij} \\ g_j & H_{ji} & H_{jj} \end{pmatrix}$ over the implicit gradient and Hessian, in the hope of a denominator-free coupling. This computes a *different* invariant. For $F = P - f$ it returns $\Lambda_P^{\text{BH}} = Ab^2 - 2abB + Ca^2$, $\Lambda_D^{\text{BH}} = C$, $\Lambda_S^{\text{BH}} = A$ —a Gauss–Kronecker-type quantity, not the mixed coupling $\Lambda_P = B$. On the keystone the bordered triple is $(\frac{1}{2}, -\frac{1}{4}, -\frac{13}{4})$ against the channel triple $(-\frac{5}{4}, \frac{1}{6}, \frac{1}{2})$; the two have different zeros. The bordered Hessian is therefore not a route to $\mathcal{A}_c$, and the projective normal (Prop. 2) and homogeneous gap vector (Prop. 3) are the correct denominator-free handles. [symbolic]

6 The role-jet groupoid and the invariant ring

On the regular locus $\mathcal{U}_{\text{role}} = \{(a, b) : ab \neq 0\}$ the role transpositions act on the full 2-jet. The first-order action is

$$s(a, b) = (b, a), \quad t(a, b) = \left(\frac{1}{a}, -\frac{b}{a}\right),$$

and on the second jet t transports

$$t(a, b, A, B, C) = \left(\frac{1}{a}, -\frac{b}{a}, -\frac{A}{a^3}, \frac{Ab - aB}{a^3}, \frac{-Ab^2 + 2abB - a^2C}{a^3}\right), \quad s(a, b, A, B, C) = (b, a, C, B, A).$$

Computational Result 5 (S_3 relations). $s^2 = t^2 = (st)^3 = e$, verified on (a, b) and (for t^2) on the full 2-jet. The role group is the genuine symmetric group S_3 , not merely a set of charts. [symbolic]

This is the structural reason the DBP local theory is well-posed where ensemble-dependent thermodynamic geometry is not: the role swaps close into a finite group with verified relations, so the three coupling curvatures form one S_3 -orbit and “role-blind” versus “role-distinguishing” is the canonical $\mathbf{1} \oplus \mathbf{2}$ decomposition rather than an arbitrary choice.

Proposition 4 (Invariant ring). *With $x_i = \kappa_{c,i}$, $\mathbb{Q}[x_1, x_2, x_3]^{S_3} = \mathbb{Q}[e_1, e_2, e_3]$ (fundamental theorem of symmetric polynomials), and $\mathcal{A}_c = 2(e_1^2 - 3e_2)$ is the lowest non-trivial role-distinction invariant. The role-blind mean e_1 is the closest analogue of a Legendre-invariant scalar; $e_3 = \prod_i \kappa_{c,i}$ is the signed (chiral) invariant. [proven]*

7 Application: the Kerr black hole

For a thermodynamic fundamental relation the three role-output charts are the standard representations: P -output (mass/energy) is the Weinhold representation, D -output (entropy) is the Ruppeiner representation, the third is the remaining-charge representation, and the transposition t is the Legendre swap. The full comparison with geometrothermodynamics—in which $\mathcal{A}_c$ is shown to be the norm of precisely the off-diagonal coupling that Legendre-invariant metrics suppress, with both pipelines carried through Kerr and the conventional GTD-II curvature verified to diverge on the Davies locus while remaining blind to the extremal boundary—is the subject of a separate companion (“Two Geometric Readings of a Thermodynamic Surface”). Here we record only what the local invariant of this paper says about Kerr.

The Kerr relation is $M = \sqrt{S/4\pi + \pi J^2/S}$, read as the graph $M = f(S, J)$ ($D = S$, substrate J).

- **Davies point** (heat capacity $C_J \propto a/A$ diverges, $A = M_{SS} = 0$) is $S = 2\pi J\sqrt{3 + 2\sqrt{3}}$, i.e. $J^2/M^4 = 2\sqrt{3} - 3 \approx 0.4641$, equivalently $48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4 = 0$; all verified exactly. [computer-verified]
- **Davies sits on the isotropy branch.** At any point with $A = f_{DD} = 0$, $\Lambda_D = (0 - aB)/a = -\Lambda_P$, hence $\kappa_{c,D} = \kappa_{c,P}$: the Weinhold and Ruppeiner coupling curvatures coincide in magnitude there, with the third representation at $\Lambda_S/\Lambda_P = -(1 + \sqrt{3})$ —partial, not full, isotropy. [symbolic]
- **Boundaries for free.** The role-singular loci are the physical boundaries: $a = M_S = T = 0$ is extremality ($S = 2\pi J$), and $b = M_J = \Omega = 0$ is Schwarzschild ($J = 0$), recovered as chart singularities of the construction. [symbolic]
- **Full isotropy is empty** in the physical wedge, as for the keystone.

Corollary 1. *The Davies locus of any DBP surface ($f_{DD} = 0$) is contained in the $|\Lambda_P| = |\Lambda_D|$ component of $\mathcal{Z}_\Sigma$. Heat-capacity divergence and energy/entropy coupling-curvature coincidence are the same stratum.*

8 Discussion and open problems

Link to the global theory. [conjecture] The cube-root monodromy phases of the companion paper live in the standard **2** of the within-block S_3 from degree-3 root extraction; the local role-coupling content $\mathcal{A}_c = 3\|P_2\kappa_c\|^2$ lives in the standard **2** of the role S_3 . Whether these are the *same* S_3 —so that global branched-cover structure and local coupling geometry are two readings of one representation—is the central open question joining the two papers.

Completing the geometric program. Whether $\mathcal{A}_c$ can be promoted from a scalar field to the curvature of a genuine DBP metric—and the related questions of an S_3 -covariant metric with boundary-degeneracy built in, a role-transport holonomy conjecturally equal to $\mathcal{A}_c$, and the interpretation of the higher invariants e_1, e_3 —is posed as a structured set of derivation targets in the GTD companion, against which the present exact $\mathcal{A}_c$ structure (flat on multiplication, nowhere flat on the keystone) is the validation data. We do not pursue them here.

Higher-order channels (unverified). [conjecture] The construction generalises to curvature order r via bordered minors of $tH_c + uH_s$, yielding a channel density $\widehat{\mathcal{C}}_r(t, u)$ and a conjectured parity law—exactness in $\mathbb{Q}$ for even r and in $\mathbb{Q}(\sqrt{q})$ for odd r —with $\mathcal{A}_c$ the $r = 2$, role-symmetric instance. These are stated as targets; none is verified here.

Convergence, not novelty. The wrapper $\mathcal{A}_c = \sum_{i < j} (x_i - x_j)^2$ is the elementary identity relating summed squared pairwise differences to variance, hence to the standard-rep projection norm; this is classical and is cited as convergence validation, not as a new result. The contribution is the *use*: reading the role variance as signal, with an exact zero locus and a denominator-free global evaluation, in a setting (DBP) whose role swaps form a genuine group.

A Verification protocol

All symbolic identities (the three forms of $\mathcal{A}_c$, the telescoping, the linear channel map and $\det = -1$, the sign branches, the S_3 relations $s^2 = t^2 = (st)^3 = e$ and full-jet $t^2 = e$, the projective recovery, the homogeneous gap vector and its telescoping, the bordered-Hessian non-equivalence) are exact computer algebra. Numerical witnesses (keystone $\mathcal{A}_c = 42793/1555848$; Kerr Davies $J^2/M^4 = 2\sqrt{3} - 3$ and the quartic $48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4 = 0$; extremal $S = 2\pi J$) are exact rationals/radicals, not floating point. The ruled-surface expression in §4 is given with the $(2DS+1)^4$ denominator restored; an earlier informal statement omitting it understated the role-singular pole and is corrected here.

Acknowledgements

Adversarial audit and exact- $\mathbb{Q}$ verification were carried out in an interactive session. The thermodynamic-geometry framing draws on the standard Weinhold/Ruppeiner/Quevedo literature, cited as context.
[To be completed.]